

OCR A A-Level Chemistry — Practical Activity Groups (PAGs)

Table of Contents

1. [PAG1: Moles Determination](#)
2. [PAG2: Acid–Base Titration](#)
3. [PAG3: Enthalpy Determination](#)
4. [PAG4: Qualitative Analysis of Ions](#)
5. [PAG5: Synthesis of an Organic Liquid](#)
6. [PAG6: Synthesis of an Organic Solid](#)
7. [PAG7: Qualitative Analysis of Organic Functional Groups](#)
8. [PAG8: Electrochemical Cells](#)
9. [PAG9: Rates of Reaction — Continuous Monitoring Method](#)
10. [PAG10: Rates of Reaction — Initial Rates Method](#)
11. [PAG11: pH Measurement](#)
12. [PAG12: Research Skills](#)

Supporting practicals (not PAGs):

- [Colorimetry and Transition Metal Complexes](#)
- [Thin-Layer Chromatography](#)
- [Investigating Equilibria](#)

OCR Practical Endorsement: the specification (H432, section 5g) defines twelve Practical Activity Groups, PAG1–PAG12. You must complete a minimum of 12 practical activities covering all of them. The experiment in each section below is one suitable example; your school may use a different activity for the same PAG.

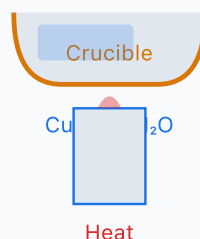
PAG1: Moles Determination

A: Molar Gas Volume



$\text{Mg} + \text{HCl} \rightarrow \text{H}_2$
Measure $V(\text{H}_2)$

B: Water of Crystallisation



Balance (weigh before and after heating)

Mass lost = mass of water. Calculate $n(\text{H}_2\text{O})$ per $n(\text{salt})$.

Molar gas volume = V/n (at room T and P) $\approx 24 \text{ dm}^3 \text{ mol}^{-1}$

Apparatus

- Gas volume:** Conical flask, gas syringe (100 cm^3), delivery tube, balance, measuring cylinder, Mg ribbon, dilute HCl
- Water of crystallisation:** Crucible, pipe-clay triangle, tripod, Bunsen burner, balance (to 0.01 g), hydrated salt (e.g., $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)

Method (A — Molar Gas Volume)

1. Weigh a known length of Mg ribbon accurately.

2. Place excess dilute HCl in a conical flask connected to a gas syringe.
3. Add the Mg ribbon, quickly seal, and collect the H₂ gas.
4. Record the final volume of gas at room temperature and pressure.
5. Calculate: $n(\text{Mg}) = \text{mass}/A_r$. Since $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$, $n(\text{H}_2) = n(\text{Mg})$.
6. Molar gas volume = $V(\text{gas}) / n(\text{H}_2)$.

Method (B — Water of Crystallisation)

1. Weigh a clean, dry crucible. Add hydrated salt and weigh again.
2. Heat gently at first, then more strongly. Allow to cool. Weigh.
3. Reheat, cool, and reweigh until constant mass is achieved (to ensure all water is driven off).
4. Mass of water = mass before - mass after. Mass of anhydrous salt = final mass - crucible mass.
5. Calculate $n(\text{water})$ and $n(\text{salt})$. The ratio gives x in the formula $\text{MX} \cdot x\text{H}_2\text{O}$.

Safety

- H₂ gas is flammable — no naked flames near gas syringe method.
- Hot crucible — use tongs; place on heatproof mat.
- Dilute HCl is irritant — goggles.

Analysis

Gas volume: Molar gas volume $\approx 24,000 \text{ cm}^3 \text{ mol}^{-1}$ at room temperature and pressure.

Water of crystallisation: Calculate the integer ratio to find x . For $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$, expect $x = 5$.

Error Analysis

Source of Error	Type	Improvement
Gas escaping before bung fitted	Systematic	Seal quickly; use excess acid
Gas syringe friction	Systematic	Tap gently; lubricate
Not heating to constant mass	Systematic	Reheat and reweigh until mass is constant

Salt spitting out of crucible	Systematic	Heat gently at first; use a lid
Reabsorption of water on cooling	Systematic	Cool in a desiccator

Exam Tip: "Heating to constant mass" means heating, cooling, weighing, reheating, and reweighing until the mass doesn't change (difference < 0.01 g). This confirms all water of crystallisation has been removed. Always cool in a desiccator to prevent reabsorption of moisture.

Worked Example — Water of Crystallisation:

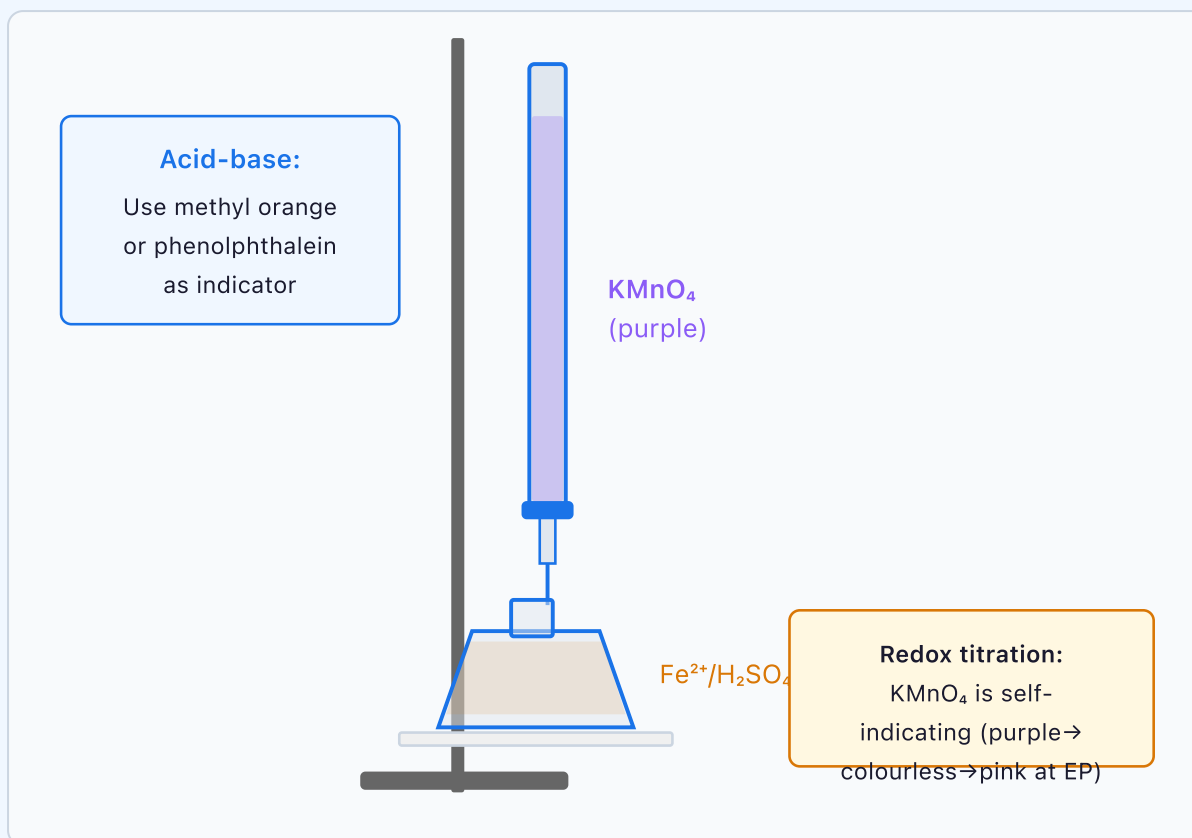
Mass of crucible = 15.20 g. Mass of crucible + hydrated salt = 20.18 g. Mass after heating to constant mass = 18.38 g.

Mass of water lost = $20.18 - 18.38 = 1.80 \text{ g} \rightarrow n(\text{H}_2\text{O}) = 1.80 / 18.0 = 0.100 \text{ mol}$

Mass of anhydrous salt = $18.38 - 15.20 = 3.18 \text{ g} \rightarrow n(\text{CuSO}_4) = 3.18 / 159.6 = 0.0199 \text{ mol}$

Ratio = $0.100 / 0.0199 = 5.02 \approx 5$, so formula is **$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$**

PAG2: Acid–Base Titration



Apparatus

- Burette (50 cm^3), pipette (25.0 cm^3), conical flasks
- White tile, indicators (for acid-base)
- **Acid-base:** 250 cm^3 volumetric flask, balance (to 0.01 g), weighing boat, wash bottle, standard acid/alkali solutions (e.g. NaHCO_3 and HCl), methyl orange or phenolphthalein
- **Redox:** KMnO_4 solution, acidified Fe^{2+} solution (Mohr's salt + H_2SO_4)

Method (Acid–Base — standard solution and titration)

1. **Make a standard solution:** weigh the solid (e.g. sodium hydrogencarbonate) accurately in a weighing boat, transfer to a beaker, reweigh the boat, and dissolve the solid in distilled water.
2. Transfer the solution to a 250 cm^3 volumetric flask using a funnel, rinsing the beaker and funnel into the flask. Make up to the graduation mark with distilled water (bottom of meniscus on the line, read at eye level). Stopper and invert several times to mix.
3. Rinse the pipette with the solution, then pipette 25.0 cm^3 into a conical flask. Add 2–3 drops of a suitable indicator.

4. Rinse and fill the burette with the other solution (e.g. HCl); remove the funnel and make sure the jet is full. Record the initial reading to 0.05 cm³.
5. Do a rough titration, then accurate titrations, adding dropwise near the end point and swirling, until you have concordant titres (within 0.10 cm³).

Choice of indicator: strong acid + strong base — either indicator; strong acid + weak base (e.g. HCl + NaHCO₃) — methyl orange (yellow → orange); weak acid + strong base — phenolphthalein (colourless → pink). Weak acid + weak base has no sharp end point.

Method (Extension — Redox Titration, KMnO₄/Fe²⁺)

Redox titrations (module 5) use the same burette and pipette technique and are examinable, but the PAG2 activity itself is an acid–base titration.

1. Pipette 25.0 cm³ of acidified Fe²⁺ solution into a conical flask.
2. Fill the burette with standardised KMnO₄ solution.
3. Add KMnO₄ slowly, swirling. The purple colour is decolourised as MnO₄[−] is reduced to Mn²⁺.
4. End point: the first permanent pale pink colour that persists for 30 seconds (no indicator needed — KMnO₄ is self-indicating).
5. Record the titre. Repeat for concordant results.
6. $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$ (1:5 ratio).

Safety

- Dilute HCl is an irritant — goggles. Use a pipette filler, never mouth pipetting.
- KMnO₄ stains skin and is an oxidiser — wear gloves.
- Sulfuric acid is corrosive — goggles.
- Use H₂SO₄ (not HCl or HNO₃) to acidify — HCl would be oxidised by KMnO₄, HNO₃ is itself an oxidiser.

Analysis

Acid–base: mean of concordant titres → $n(\text{titrant}) = c \times V/1000$ → use the balanced equation (NaHCO₃ + HCl → NaCl + H₂O + CO₂ is 1:1) → concentration of the other solution = n/V .

Redox — calculate: $n(\text{KMnO}_4) = c \times V(\text{titre})$. $n(\text{Fe}^{2+}) = 5 \times n(\text{KMnO}_4)$. Then $[\text{Fe}^{2+}] = n/V$.

Error Analysis

Source of Error	Type	Improvement
Solid not fully transferred / flask not made up exactly to the mark	Systematic	Rinse beaker, rod and funnel into the flask; read the meniscus at eye level
Air bubble in burette jet or funnel left in burette	Systematic	Run some solution through the jet first; remove the funnel before reading
Overshooting end point	Systematic	Add drop-wise near end; swirl between drops
KMnO ₄ staining burette (hard to read)	Systematic	Rinse well; read from top of meniscus (KMnO ₄ is opaque)
Not using H ₂ SO ₄ as the acid	Systematic	Always use sulfuric acid for KMnO ₄ titrations
Fe ²⁺ oxidised by air before titration	Systematic	Use freshly prepared solution; keep acidified

Exam Tip: For KMnO₄ titrations, read the burette from the TOP of the meniscus (not the bottom) because the solution is so dark you cannot see through it. No indicator is needed — KMnO₄ is self-indicating. The end point is a permanent faint pink colour.

Worked Example — Redox Titration:

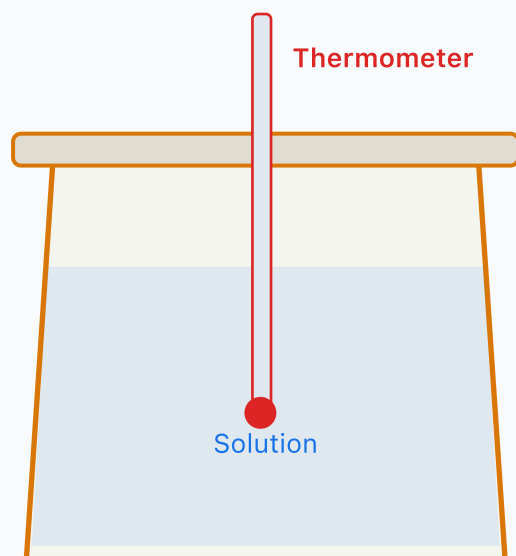
25.0 cm³ of Fe²⁺ solution required 23.60 cm³ of 0.0200 mol dm⁻³ KMnO₄.

$$n(\text{KMnO}_4) = 0.0200 \times 23.60/1000 = 4.72 \times 10^{-4} \text{ mol}$$

$$\text{From equation: } n(\text{Fe}^{2+}) = 5 \times n(\text{KMnO}_4) = 5 \times 4.72 \times 10^{-4} = 2.36 \times 10^{-3} \text{ mol}$$

$$[\text{Fe}^{2+}] = 2.36 \times 10^{-3} / 0.0250 = \mathbf{0.0944 \text{ mol dm}^{-3}}$$

PAG3: Enthalpy Determination



Hess's Law

$$\Delta H_1 = \Delta H_2 + \Delta H_3$$

Measure ΔH_2 and ΔH_3

to find ΔH_1

$$q = mc\Delta T \mid \Delta H = -q/n$$

Use graphical extrapolation for accurate ΔT

Apparatus

- Polystyrene cup with lid
- Thermometer ($\pm 0.1^\circ\text{C}$) or data logger
- Measuring cylinder or pipette
- Stopwatch
- Reagents for Hess's law cycle (e.g., $\text{NaOH} + \text{HCl}$ direct, and via $\text{NaOH} + \text{NH}_4\text{Cl}$ then $\text{NH}_3 + \text{HCl}$)

Method

1. Measure a known volume of solution into the polystyrene cup.
2. Record temperature every minute for 3 minutes (baseline).
3. Add the second reagent at the 4th minute. Stir and record temperature for 6+ more minutes.
4. Plot temperature vs. time. Extrapolate to find the true ΔT .

5. Calculate $q = mc\Delta T$. Then $\Delta H = -q/n$.

6. **For Hess's Law:** Perform two separate experiments to measure two enthalpy changes in a cycle. Use Hess's Law to calculate the third (unmeasurable) enthalpy change.

Safety

- Acids and alkalis — corrosive, wear goggles.
- Exothermic reactions — cup may become warm.

Analysis

Plot: Temperature vs. time with graphical extrapolation.

Hess's Law: $\Delta H_1 = \Delta H_2 + \Delta H_3$ (or appropriate cycle). Compare calculated value with literature.

Error Analysis

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic	Lid, insulation, graphical extrapolation
Assuming $c = 4.18$ and density = 1	Systematic	Acknowledged approximation
Heat absorbed by calorimeter	Systematic	Determine calorimeter constant
Incomplete mixing	Random	Stir thoroughly; use magnetic stirrer

Exam Tip: Hess's Law: enthalpy change for a reaction is independent of the route taken, provided initial and final conditions are the same. You can measure ΔH (formation) indirectly by measuring ΔH (combustion) of reactants and products. Draw a clear energy cycle in your answer. Always use graphical extrapolation of your temperature-time plot to correct for heat loss — this significantly improves accuracy.

PAG4: Qualitative Analysis of Ions

The diagram illustrates various qualitative analysis tests for ions. It features three test tubes on the left, each containing a precipitate: a blue precipitate for Cu^{2+} , a green precipitate for Fe^{2+} , and a brown precipitate for Fe^{3+} . Below these are labels: 'Blue ppt', 'Green ppt', and 'Brown ppt'. To the right of the test tubes is a Bunsen burner with a red flame, labeled 'Flame tests' above it and 'Li Na K Ca Ba' below it. Further right is a box titled 'Gas tests' containing the following information: CO_2 : limewater, Cl_2 : litmus, NH_3 : damp red litmus $\rightarrow$ blue. At the bottom of the diagram, a line of text lists anion tests: 'Anions: $\text{Cl}^-/\text{Br}^-/\text{I}^-$ (AgNO_3) | SO_4^{2-} (BaCl_2) | CO_3^{2-} (HCl + limewater)'. The entire diagram is set against a light blue background with a white border.

Cu^{2+} Fe^{2+} Fe^{3+} Flame tests

Blue ppt Green ppt Brown ppt Li Na K Ca Ba

Gas tests

CO_2 : limewater
 Cl_2 : litmus
 NH_3 : damp red litmus $\rightarrow$ blue

Anions: $\text{Cl}^-/\text{Br}^-/\text{I}^-$ (AgNO_3) | SO_4^{2-} (BaCl_2) | CO_3^{2-} (HCl + limewater)

Apparatus

- Test tubes, nichrome wire, Bunsen burner
- $\text{NaOH}(\text{aq})$, $\text{NH}_3(\text{aq})$, $\text{AgNO}_3(\text{aq})$, $\text{BaCl}_2(\text{aq})$ or $\text{Ba}(\text{NO}_3)_2(\text{aq})$
- Dilute HNO_3 , dilute HCl
- Limewater, litmus paper

Method

1. **Order of anion tests (OCR):** carbonate, then sulfate, then halide. BaCO_3 is insoluble and Ag_2SO_4 is only slightly soluble (both can precipitate), so doing the tests in this order on the same sample avoids false positives.
2. **Carbonate test:** Add dilute HNO_3 . Effervescence; gas turns limewater milky = CO_3^{2-} . Keep adding acid until fizzing stops.
3. **Sulfate test:** To the acidified solution add $\text{Ba}^{2+}(\text{aq})$. White ppt (BaSO_4) = SO_4^{2-} . Use $\text{Ba}(\text{NO}_3)_2$ rather than BaCl_2 if you will test the same solution for halides.
4. **Halide tests:** Acidify with HNO_3 , add AgNO_3 . White (AgCl), cream (AgBr), yellow (AgI) precipitate. Confirm with $\text{NH}_3(\text{aq})$: AgCl dissolves in dilute NH_3 , AgBr only in concentrated

NH_3 , AgI is insoluble in both.

5. **Ammonium test:** Add $\text{NaOH}(\text{aq})$, warm gently. Test gas with damp red litmus — turns blue = NH_4^+ .

6. **Transition metal cations (module 5):** Add $\text{NaOH}(\text{aq})$ or $\text{NH}_3(\text{aq})$ dropwise, then in excess. Cu^{2+} pale blue ppt (dissolves in excess NH_3 to a deep blue solution); Fe^{2+} green ppt; Fe^{3+} orange-brown ppt; Mn^{2+} light brown ppt; Cr^{3+} grey-green ppt (dissolves in excess NaOH to a dark green solution and in excess NH_3 to a purple solution).

7. **Flame tests** (useful background, not required by the OCR A specification): clean nichrome wire in HCl , dip in sample, observe colour.

Safety

- AgNO_3 stains — gloves. BaCl_2 is toxic.
- Conc. HCl in fume cupboard.
- Goggles throughout.

Analysis

Record: Systematic table of observations for each test and each unknown ion.

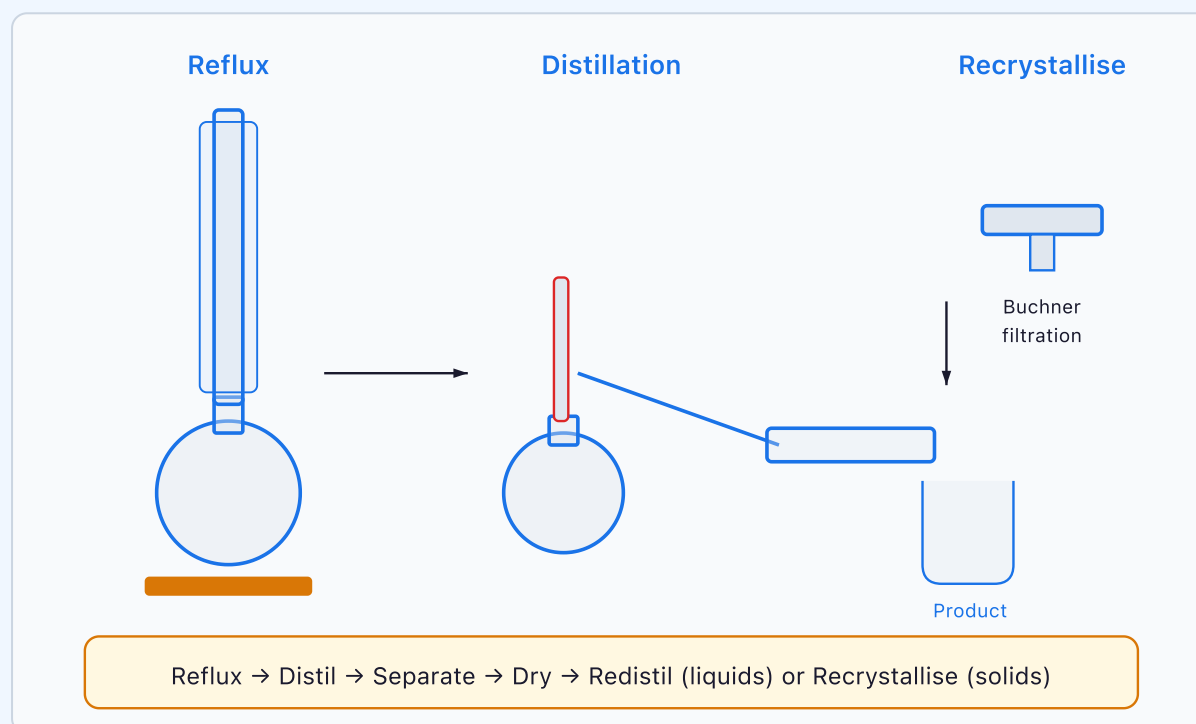
Error Analysis

Source of Error	Type	Improvement
Contaminated wire/glassware	Systematic	Clean thoroughly before each test
Tests done in the wrong order (false positives)	Systematic	Test carbonate → sulfate → halide
Not acidifying before $\text{AgNO}_3/\text{BaCl}_2$	Systematic	Always acidify with HNO_3 first
Similar precipitate colours	Random	Compare with known standards

Exam Tip: $\text{Cr}(\text{OH})_3$ dissolves in excess NaOH (dark green solution) and in excess NH_3 (purple $[\text{Cr}(\text{NH}_3)_6]^{3+}$). $\text{Cu}(\text{OH})_2$ dissolves in excess NH_3 (deep blue $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$) but

not in excess NaOH. These excess-reagent observations are the key distinguishing tests in exams. Always acidify with dilute HNO_3 before adding AgNO_3 (halide test) or BaCl_2 (sulfate test) to prevent false positives from carbonates or sulfites.

PAG5: Synthesis of an Organic Liquid



Apparatus

- Round-bottomed flask, anti-bumping granules
- Reflux condenser (vertical), distillation condenser (angled)
- Separating funnel
- Drying agent (anhydrous MgSO_4 , Na_2SO_4 or CaCl_2)
- Thermometer for the still head; electric heating mantle
- Example reagents (haloalkane): butan-1-ol, sodium bromide, concentrated sulfuric acid, sodium hydrogencarbonate solution

Method (e.g. 1-bromobutane from butan-1-ol)

1. **Reflux:** Heat butan-1-ol with sodium bromide and concentrated sulfuric acid (which form HBr in situ) in a round-bottomed flask with anti-bumping granules and a vertical condenser. Reflux lets the reaction go to completion without losing volatile reactants or product.
2. **Distillation:** Rearrange for distillation and collect the crude liquid product.
3. **Separating funnel:** Wash the distillate with water, then with sodium hydrogencarbonate solution (removes acid; vent often, CO₂ is released), then water again. Keep the organic layer — 1-bromobutane is denser than water so it is the *lower* layer.
4. **Dry:** Add anhydrous drying agent and swirl until the liquid is clear, then decant or filter.
5. **Redistil:** Collect the fraction boiling close to the data-book value (about 100–104 °C for 1-bromobutane).
6. Weigh the product and calculate the percentage yield.

Safety

- Flammable organics — no naked flames; electric heating.
- Fume cupboard for volatile substances.
- Never heat sealed apparatus.
- Vent separating funnel to release pressure.
- Concentrated sulfuric acid is corrosive — add slowly with cooling. A written risk assessment is part of this PAG.

Analysis

Calculate: Percentage yield = (actual/theoretical) × 100. Atom economy = (M_r of desired product / sum of M_r of all products) × 100.

Purity: the product distils over a narrow range at the data-book boiling point. A clear (not cloudy) liquid shows it is dry.

Error Analysis

Source of Error	Type	Improvement
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Product lost during transfers	Systematic	Minimise transfers; rinse equipment
Incomplete reaction	Systematic	Reflux for longer; use excess reagent
Impure product (distils over a wide range)	Systematic	Redistil, collecting only a narrow fraction
Wrong layer kept in separating funnel	Systematic	Check densities; add a drop of water to see which layer it joins
Wet product (cloudy)	Systematic	Add more drying agent until the liquid is clear

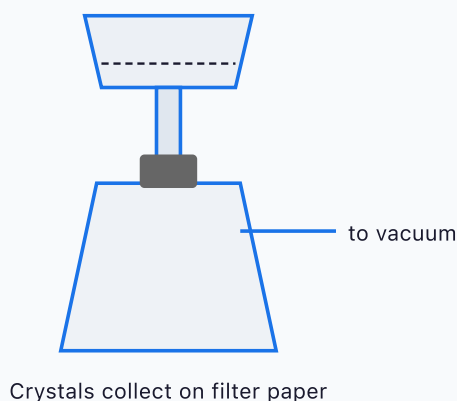
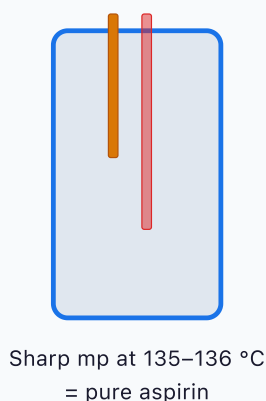
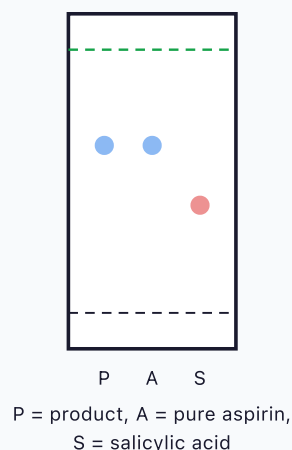
Exam Tip: Know why each step is done: *reflux* (heat for a long time without losing volatile substances), *distillation* (separate the volatile product), *washing with NaHCO₃* (neutralise acidic impurities), *drying agent* (remove water), *redistillation* (purify; collect at the boiling point). Never heat a sealed system — the top of the condenser is left open. Yield-reducing factors: incomplete reaction, side reactions, losses during transfers and in the separating funnel.

Worked Example — Percentage Yield:

7.40 g of butan-1-ol ($M = 74.0 \text{ g mol}^{-1}$) gives $n = 0.100 \text{ mol}$. Theoretical mass of 1-bromobutane ($M = 137.0 \text{ g mol}^{-1}$) = $0.100 \times 137.0 = 13.7 \text{ g}$. Actual mass after redistillation = 8.22 g.

Percentage yield = $(8.22 / 13.7) \times 100 = 60.0\%$

PAG6: Synthesis of an Organic Solid

Buchner filtration**Melting point****TLC check**

React → Crystallise → Filter → Recrystallise → Dry → Melting point & TLC

Apparatus

- Conical flask or pear-shaped flask (with reflux condenser if heating for longer), water bath or electric heater
- Salicylic acid (2-hydroxybenzoic acid), ethanoic anhydride, a few drops of concentrated phosphoric(V) or sulfuric acid as catalyst
- Buchner funnel, filter flask, filter paper, vacuum pump; ice bath
- Solvent for recrystallisation (e.g. hot water or an ethanol–water mixture)
- Melting point apparatus and capillary tubes
- TLC plate, developing chamber, capillary spotters, UV lamp

Method (Synthesis of Aspirin)

1. Weigh about 2 g of salicylic acid accurately into a dry flask. Add excess ethanoic anhydride (about 4 cm³) in a fume cupboard, then a few drops of the acid catalyst.
2. Warm in a water bath at roughly 80–90 °C for 10–15 minutes, swirling, until all the solid has dissolved. (Heating under reflux can be used instead; reflux may be covered in either PAG5 or PAG6.)
3. Carefully add cold water to hydrolyse the excess ethanoic anhydride, then cool the flask in an ice bath to crystallise the crude aspirin.

4. Collect the crystals by suction filtration using a Buchner funnel. Wash with a little ice-cold water.
5. **Recrystallise:** dissolve the crude solid in the minimum volume of hot solvent, filter while hot if there are insoluble impurities, then cool slowly so that pure crystals form and soluble impurities stay in solution. Filter again under reduced pressure, wash with a little cold solvent, and dry (e.g. between filter papers or in a desiccator).
6. Weigh the dry product and calculate the percentage yield.
7. **Check purity:** measure the melting point range and compare with the data-book value (about 136 °C). Run a TLC plate spotted with the product, pure aspirin and salicylic acid.

Safety

- Ethanoic anhydride is corrosive and its vapour is irritating to the eyes and lungs — dispense in a fume cupboard.
- Concentrated phosphoric(V)/sulfuric acid is corrosive — goggles and gloves.
- Ethanol (if used in recrystallisation) is flammable — heat with a water bath or electric heater, not a flame.
- A written risk assessment is part of this PAG: identify each hazard, the risk and how it is controlled.

Analysis

Calculate: Percentage yield = (actual mass / theoretical mass) × 100, using the limiting reagent (salicylic acid; ethanoic anhydride is in excess).

Purity: A pure solid melts sharply (over 1–2 °C) at the data-book value. On the TLC plate, a pure product gives a single spot at the same R_f as pure aspirin; a spot matching salicylic acid shows unreacted starting material.

Error Analysis

Source of Error	Type	Improvement
Too much solvent used in recrystallisation	Systematic	Use the minimum volume of hot solvent so less product stays dissolved

Product lost during transfers and filtration	Systematic	Minimise transfers; rinse with small amounts of ice-cold solvent
Wet product (mass too high; mp lowered)	Systematic	Dry thoroughly; dry to constant mass
Heating melting point apparatus too quickly	Systematic	Heat slowly (about 1–2 °C per minute) near the expected mp

Exam Tip: Be ready to explain each recrystallisation step: *minimum* hot solvent (so the product crystallises on cooling), hot filtration (removes insoluble impurities), slow cooling (purer, larger crystals), cold-solvent wash (removes soluble impurities on the crystal surface without redissolving much product). Impurities *lower and broaden* the melting point range.

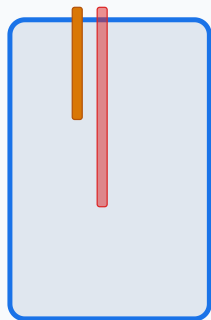
Worked Example — Percentage Yield:

A student prepared aspirin from 2.00 g of salicylic acid. The theoretical yield is 2.61 g. The actual yield after purification was 1.85 g.

$$\text{Percentage yield} = (1.85 / 2.61) \times 100 = \mathbf{70.9\%}$$

This is a typical yield for a lab preparation. Losses occur during transfer, filtration, washing, and recrystallisation.

PAG7: Qualitative Analysis of Organic Functional Groups



Melting point
apparatus

Lab Tests

$\text{Br}_2(\text{aq}) \rightarrow \text{C}=\text{C}$
 $2,4\text{-DNP} \rightarrow \text{C}=\text{O}$
Tollens' $\rightarrow \text{CHO}$
 $\text{Na}_2\text{CO}_3 \rightarrow \text{COOH}$
 $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+ \rightarrow \text{OH}$
 $\text{AgNO}_3/\text{ethanol} \rightarrow \text{C-X}$

Spectroscopy

Mass spec $\rightarrow M_r$
IR $\rightarrow$ functional groups
(O-H, C=O, N-H etc.)
 ^1H NMR $\rightarrow$ H environments
 ^{13}C NMR $\rightarrow$ C environments
Splitting, integration, δ

Apparatus

- Melting point apparatus, capillary tubes
- Test tubes, water bath
- Reagents: $\text{Br}_2(\text{aq})$, $\text{AgNO}_3(\text{aq})$ in ethanol, 2,4-DNP, Tollens' reagent, acidified $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{Na}_2\text{CO}_3(\text{aq})$, limewater
- Spectroscopic data (mass spec, IR, NMR — provided in exams; not part of the PAG itself)

Method

1. Perform the functional group tests systematically on small samples in test tubes (see the table below), heating in a water bath where needed.
2. Record all positive and negative results in a table.
3. If the unknown is a solid, its melting point can be compared with data-book values.
4. Combine lab results with spectroscopic data to identify the compound:
 - Mass spectrum: molecular ion peak gives M_r ; fragmentation pattern gives structural clues.
 - IR spectrum: broad O-H at $2500\text{--}3300\text{ cm}^{-1}$ (carboxylic acid), sharp C=O at $\sim 1700\text{ cm}^{-1}$, broad O-H at $3200\text{--}3600\text{ cm}^{-1}$ (alcohol).
 - NMR: number of peaks = number of unique H/C environments. Integration ratio = relative number of H. Splitting pattern (n+1 rule).

Functional group	Test	Positive result
Alkene (C=C)	Shake with bromine water	Orange → colourless
Haloalkane	Warm with AgNO ₃ (aq) in ethanol (water bath)	Precipitate: white (Cl), cream (Br), yellow (I); C–I fastest, C–Cl slowest
Phenol	Test pH; add Na ₂ CO ₃ (aq)	Weakly acidic but <i>no</i> effervescence with carbonate
Carbonyl (aldehyde or ketone)	2,4-DNP (Brady's reagent)	Orange precipitate
Aldehyde	Warm with Tollens' reagent	Silver mirror (ketones: no change)
Primary/secondary alcohol or aldehyde	Warm with acidified K ₂ Cr ₂ O ₇	Orange → green (tertiary alcohols and ketones: stays orange)
Carboxylic acid	Add Na ₂ CO ₃ (aq)	Effervescence; CO ₂ turns limewater milky

Safety

- Ethanol and many organic unknowns are flammable — heat with a water bath, no naked flames.
- K₂Cr₂O₇ is carcinogenic — gloves, fume cupboard.
- Tollens' — prepare fresh, dispose of immediately (can form explosive compounds on standing).

Analysis

Deduce: Use the pattern of positive and negative tests to identify the functional groups present (e.g. 2,4-DNP positive but Tollens' negative = ketone).

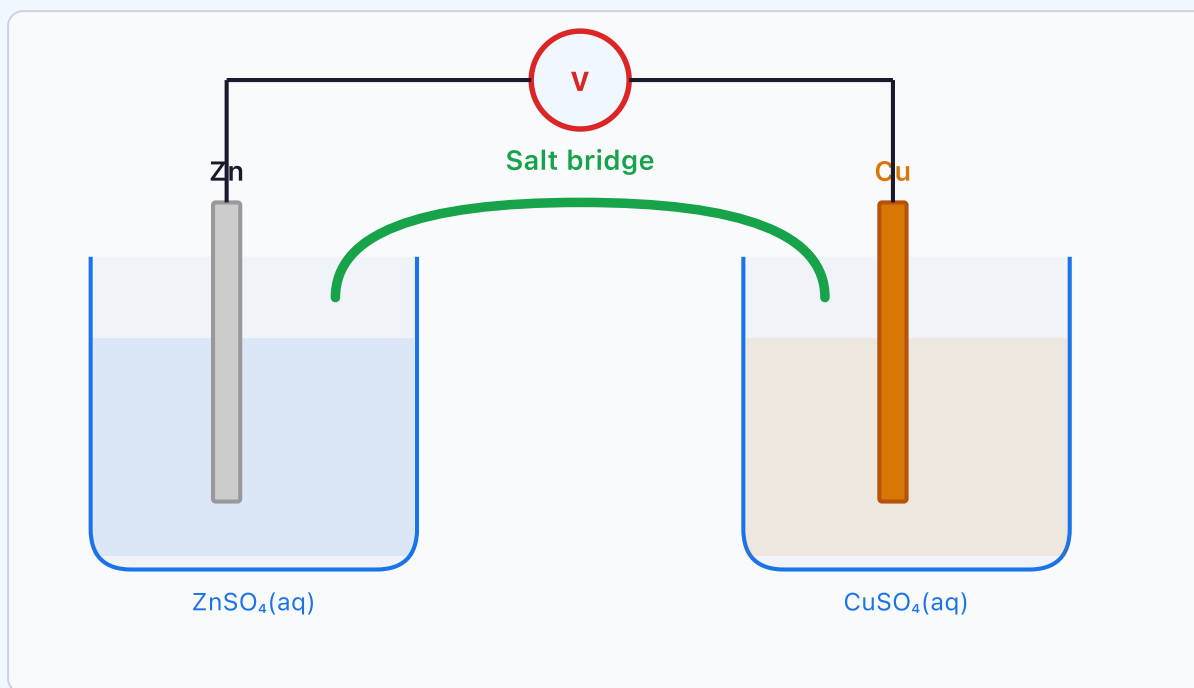
Combine (exam questions): Melting point, functional group tests, mass spec (M_r), IR (bonds), NMR (structure) to fully identify the compound.

Error Analysis

Source of Error	Type	Improvement
Impure sample (broad mp)	Systematic	Purify first by recrystallisation
False positive/negative in tests	Systematic	Run positive and negative controls alongside
Heating too fast for melting point	Systematic	Heat slowly near expected mp
Contaminated NMR solvent	Systematic	Use pure deuterated solvent

Exam Tip: In combined analytical questions, start with the mass spectrum (molecular ion gives M_r and molecular formula), then check IR for key functional groups, and finally use NMR to determine the exact structure. The number of ^1H NMR peaks tells you the number of different hydrogen environments; integration gives the ratio; splitting ($n+1$ rule) tells you adjacent hydrogens. Always calculate the degree of unsaturation (IHD) from the molecular formula — $\text{IHD} = 4$ strongly suggests a benzene ring.

PAG8: Electrochemical Cells



Apparatus

- Beakers, metal electrodes (Zn, Cu, Fe, Mg, etc.)
- 1.0 mol dm^{-3} metal salt solutions
- Salt bridge (KNO_3), high-resistance voltmeter
- Connecting wires, crocodile clips, emery paper

Method

1. Clean electrodes with emery paper.
2. Set up half-cells in separate beakers with their metal ion solutions.
3. Connect with a salt bridge and wires to voltmeter.
4. Record EMF and polarity.
5. Investigate: different metal pairs, effect of concentration (e.g. dilute the Cu^{2+} solution in a Zn/Cu cell and record the new cell potential), effect of temperature.
6. Compare with $E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode})$.

Safety

- Metal salt solutions — wear gloves and goggles.

- Dispose of solutions correctly.

Analysis

Calculate: $E^{\circ}(\text{cell}) = E^{\circ}(\text{right}) - E^{\circ}(\text{left})$. Compare measured and theoretical.

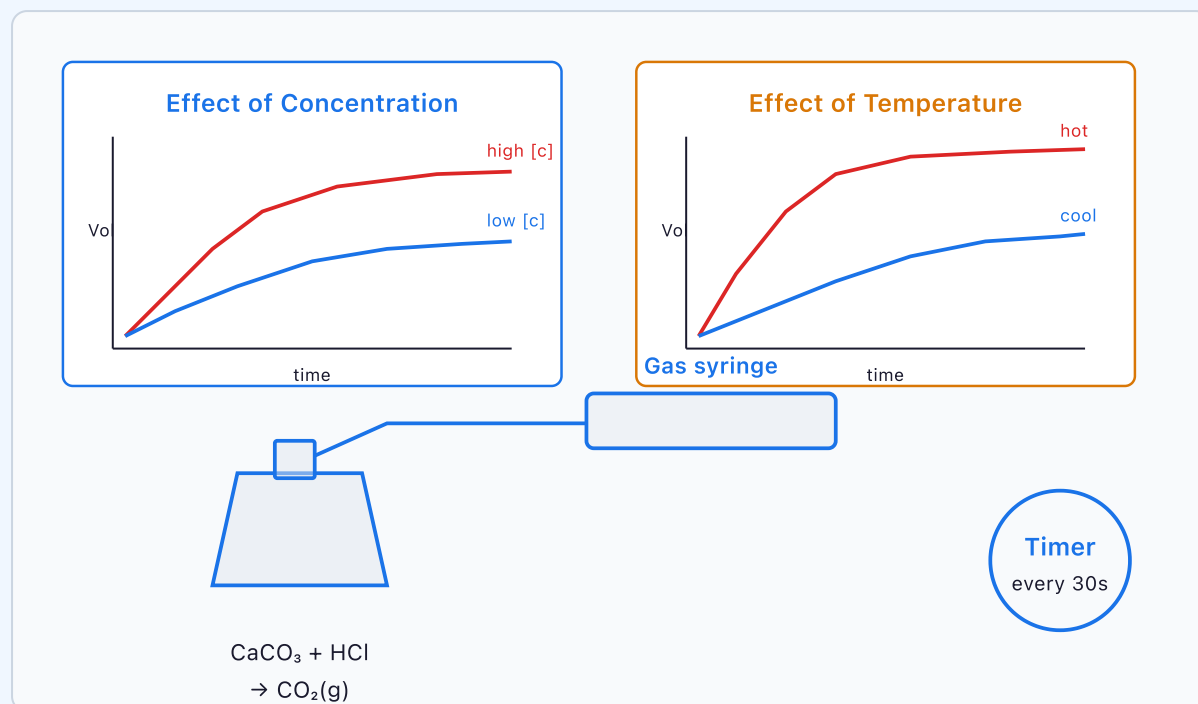
Investigate: How changing [ion] affects EMF (Nernst equation qualitatively).

Error Analysis

Source of Error	Type	Improvement
Oxide on electrodes	Systematic	Clean immediately before use
Non-standard conditions	Systematic	Record T; use 1.0 mol dm^{-3} solutions
Salt bridge resistance	Systematic	Fresh salt bridge; good contact
Voltmeter drawing current	Systematic	High-resistance voltmeter

Exam Tip: Electrons flow from the more negative electrode (anode, oxidation) to the more positive electrode (cathode, reduction) through the external circuit. Ions flow through the salt bridge to maintain electrical neutrality. The salt bridge does not allow electron flow. Remember: **AN OX** (anode = oxidation), **RED CAT** (reduction = cathode).

PAG9: Rates of Reaction — Continuous Monitoring Method



Apparatus

- Conical flask, gas syringe, delivery tube, bung
- Stopwatch
- Measuring cylinders
- CaCO_3 (marble chips) and dilute HCl (for concentration investigation)
- Alternatives: top-pan balance (mass of CO_2 lost) or a colorimeter/data logger (colour change)
- Spreadsheet software to process the data

Method

1. **Continuous monitoring:** Set up flask + gas syringe. Add the marble chips to the HCl, seal immediately and start the stopwatch. Record the volume of CO_2 every 30 s until no more gas is produced.
2. Alternatively, stand the flask (with a cotton wool plug) on a balance and record the mass every 30 s, or follow a coloured reactant/product with a colorimeter.
3. To study concentration, repeat with different concentrations of HCl (same volume, same mass and size of marble chips in excess, same temperature).
4. Plot the measured quantity against time for each run.

Safety

- HCl is irritant — goggles.
- Acid spray from the flask — cotton wool plug if measuring mass loss.

Analysis

Rate: Draw tangents to the volume–time curve; the gradient of a tangent is the rate at that time (at $t = 0$ it is the initial rate).

Half-life: The amount of reactant left is proportional to $(V_{\text{final}} - V)$. Convert to a concentration–time graph and read off successive half-lives. Constant half-life = first order; then $k = \ln 2 / t_{1/2}$.

Order from a concentration–time graph: straight line (constant gradient) = zero order; curve with constant half-life = first order.

Error Analysis

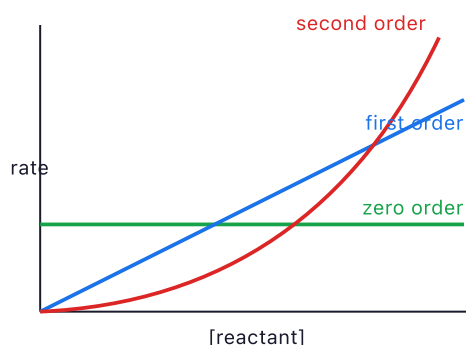
Source of Error	Type	Improvement
Gas escaping before bung fitted	Systematic	Seal quickly
Syringe reading taken late / syringe sticks	Random	Two people (one timing, one reading); use a data logger
Temperature not maintained	Systematic	Use thermostatted water bath
Uneven marble chip sizes	Random	Use similar size/mass

Exam Tip: When drawing tangents on a volume–time curve, the tangent at $t = 0$ gives the initial rate. If CaCO_3 is in excess and the same volume of HCl is used, the HCl is the limiting reagent: a more concentrated acid gives a faster initial rate *and* a larger final volume of CO_2 . Only if the acid is in excess (and the same mass of CaCO_3 is used) is the final volume the same. For a first-order reaction, the half-life is constant and independent of the starting concentration.

Worked Example — Half-life:

A concentration–time graph shows $[A]$ falling from 0.080 to $0.040 \text{ mol dm}^{-3}$ in 40 s and from 0.040 to $0.020 \text{ mol dm}^{-3}$ in a further 40 s . The half-life is constant, so the reaction is **first order** in A.

$$k = \ln 2 / t_{1/2} = 0.693 / 40 = \mathbf{0.0173 \text{ s}^{-1}}$$

PAG10: Rates of Reaction — Initial Rates Method**Iodine clock****Initial rate vs concentration**

Change ONE concentration at a time; keep total volume and temperature constant

$$\text{rate} = k[A]^n[B]^m \rightarrow k = \text{rate} / ([A]^n[B]^m)$$

Apparatus

- Conical flasks or boiling tubes, burettes or graduated pipettes for each solution
- Stopwatch; thermometer; water baths (for the temperature extension)
- **Iodine clock:** potassium iodide, dilute sulfuric acid, hydrogen peroxide, sodium thiosulfate (small fixed amount), starch solution, distilled water

- **Temperature extension:** sodium thiosulfate and dilute hydrochloric acid, paper marked with a cross
- Spreadsheet software to process the results

Method

1. Mix fixed volumes of KI(aq) , $\text{H}_2\text{SO}_4\text{(aq)}$, $\text{Na}_2\text{S}_2\text{O}_3\text{(aq)}$ and starch in a flask. Add the $\text{H}_2\text{O}_2\text{(aq)}$ last and start the stopwatch immediately.
2. The iodine formed ($\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow \text{I}_2 + 2\text{H}_2\text{O}$) is immediately turned back into I^- by the thiosulfate. When all the thiosulfate has been used, free I_2 appears and the starch suddenly turns blue-black. Record this time, t .
3. Repeat with different volumes of KI(aq) , making up the difference with distilled water so that the **total volume stays the same** and only $[\text{I}^-]$ changes.
4. Repeat the series varying $[\text{H}_2\text{O}_2]$ and then $[\text{H}^+]$ in the same way, each time keeping the other concentrations fixed.
5. Carry out each run at the same temperature (measure it) and repeat runs to check reliability.
6. **Temperature extension:** time the disappearance of a cross under a flask of $\text{Na}_2\text{S}_2\text{O}_3\text{(aq)}$ + HCl(aq) at several temperatures, keeping all volumes and concentrations the same.

Safety

- Dilute sulfuric acid and hydrochloric acid are irritants — goggles.
- Hydrogen peroxide is an irritant/oxidiser — avoid skin contact.
- The thiosulfate/acid reaction releases SO_2 — work in a well-ventilated room and dispose of the mixture promptly; take care if you have asthma.
- Hot water baths — care with scalds.

Analysis

Orders: Because the same small amount of I_2 is formed before the colour appears, the initial rate $\propto 1/t$. Plot $1/t$ against the concentration you varied: a horizontal line = zero order, a straight line through the origin = first order, an upward curve = second order (confirm by plotting $1/t$ against concentration²). Alternatively plot $\log(\text{rate})$ against $\log(\text{concentration})$: the gradient is the order.

Rate constant: Once the orders are known, substitute an initial rate and the concentrations into the rate equation to find k , and work out its units.

Temperature: Plot $\ln(1/t)$ ($\propto \ln k$) against $1/T$ (T in K). The gradient = $-E_a/R$, so $E_a = -\text{gradient} \times 8.314$.

Error Analysis

Source of Error	Type	Improvement
Judging the exact moment of colour change	Random	Same observer; white background; repeat and average; use a colorimeter/light sensor
Delay between mixing and starting the clock	Random	Add the last reagent quickly and start timing at the same moment
Total volume not kept constant	Systematic	Make up each run to the same total volume with distilled water
Temperature varies between runs	Systematic	Use a thermostatted water bath; pre-equilibrate the solutions

Exam Tip: $1/t$ is only a fair measure of the *initial* rate if the reaction has gone only a small way when the colour appears — that is why only a small, fixed amount of thiosulfate is used. Always state that you keep the concentrations of the other reactants, the total volume and the temperature constant. The units of k depend on the overall order: for rate = $k[A][B]$, k has units $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$.

Worked Example — Finding an Order:

Doubling $[I^-]$ (all else constant) changes the time from 84 s to 42 s, so $1/t$ (the initial rate) doubles → the reaction is **first order with respect to I^-** .

Doubling $[H^+]$ leaves the time unchanged at 84 s → **zero order with respect to H^+** (H^+ does not appear in the rate equation).

PAG11: pH Measurement

Buffer action test:

Add small amounts of
HCl or NaOH → pH
barely changes

Control: water

Same amount of HCl/
NaOH → pH changes
dramatically

pH 4.75

pH meter

Buffer solution

$CH_3COOH + CH_3COONa$

$pH = pK_a + \log([A^-]/[HA])$ | At half-equivalence: $pH = pK_a$

Apparatus

- Calibrated pH meter or pH probe
- Buffer solutions: weak acid (e.g., CH_3COOH) + its conjugate base (CH_3COONa)
- Dilute HCl and NaOH (for testing buffer action)
- Measuring cylinders, beakers
- Distilled water (as control)
- Unknown solutions of equal concentration (e.g. $0.100 \text{ mol dm}^{-3}$ strong acid, weak acid, strong base, weak base, neutral salt)

Method (A — Identifying Unknown Solutions)

1. Calibrate the pH meter using standard buffer solutions (e.g. pH 4, 7 and 10). Rinse the probe with distilled water between readings.
2. Measure the pH of each unknown solution (all the same concentration) and record it to 2 decimal places.
3. Identify each solution: at $0.100 \text{ mol dm}^{-3}$ a strong acid gives pH 1.0, a weak acid roughly 2–4, a neutral salt about 7, a weak base roughly 10–12 and a strong base 13.0.
4. For a weak acid, use $[\text{H}^+] = 10^{-\text{pH}}$ and $K_a = [\text{H}^+]^2/[\text{HA}]$ to estimate K_a .

Method (B — Buffer Solutions)

1. Prepare a buffer solution by mixing known volumes of weak acid and its sodium salt.
2. Measure the initial pH of the buffer.
3. Add a small measured volume of dilute HCl. Record the new pH.
4. Prepare a fresh buffer and add the same volume of NaOH. Record the pH.
5. Repeat steps 3–4 with distilled water as a control.
6. Compare: the buffer pH barely changes, while the water pH changes dramatically.
7. **Extension:** Prepare buffers with different ratios of $[\text{HA}]:[\text{A}^-]$ and verify that $\text{pH} = \text{p}K_a + \log([\text{A}^-]/[\text{HA}])$.

Safety

- Weak acids are irritants — goggles.
- NaOH is corrosive — gloves.
- pH probes are fragile — handle carefully.

Analysis

Unknowns: e.g. $0.100 \text{ mol dm}^{-3}$ ethanoic acid with pH 2.88: $[\text{H}^+] = 1.32 \times 10^{-3} \text{ mol dm}^{-3}$, $K_a = (1.32 \times 10^{-3})^2 / 0.100 = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$.

Buffers — calculate: Expected pH using Henderson-Hasselbalch: $\text{pH} = \text{p}K_a + \log([\text{A}^-]/[\text{HA}])$. Compare with measured pH.

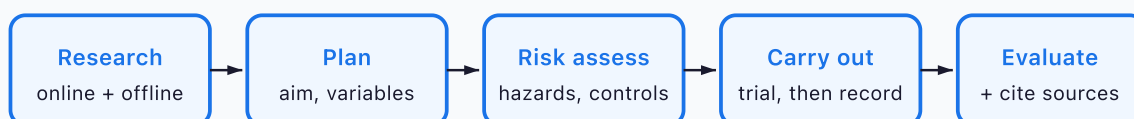
Buffer capacity: Number of moles of H^+/OH^- needed to change pH by 1 unit.

Error Analysis

Source of Error	Type	Improvement
pH meter not calibrated	Systematic	Calibrate with at least 2 standard buffers
Inaccurate volumes of HA and A ⁻	Random	Use pipettes for precise volumes
CO ₂ from air dissolving in solutions	Systematic	Use freshly prepared solutions; work quickly
Temperature effects on pH	Systematic	Record temperature; work at constant T

Exam Tip: A buffer resists pH change because: when H⁺ is added, A⁻ reacts with it ($A^- + H^+ \rightarrow HA$); when OH⁻ is added, HA reacts with it ($HA + OH^- \rightarrow A^- + H_2O$). The buffer works best when $[HA] \approx [A^-]$ (i.e., $pH \approx pK_a$). Maximum buffer capacity is at the half-equivalence point of a titration.

PAG12: Research Skills



Example question (OCR's suggestion)

How long does it take an iron tablet to break down in the stomach?
Model the stomach with dilute HCl at 37 °C; measure Fe²⁺ released over time

PAG12 is not a single fixed experiment. It is an open-ended investigation in which you research a question yourself, plan and carry out your own method, and correctly reference the sources you

used. It covers the CPAC competency on using online and offline research skills and citing sources.

Apparatus

- Depends on your plan. For the iron tablet example: iron(II) sulfate tablets, dilute HCl (to model stomach acid, pH about 1–2), a water bath at 37 °C, stirrer, stopwatch, pipettes
- For analysis of Fe^{2+} : burette and standard $\text{KMnO}_4(\text{aq})$ for redox titration, or a colorimeter with a suitable colour-forming reagent
- Access to textbooks, journals, reputable websites and data books for research

Method (Investigative Approach)

1. **Research:** find background information and possible methods from several sources (e.g. textbooks, the CLEAPSS/RSC resources your school uses, scientific articles). Note full details of each source as you go.
2. **Plan:** write an aim and a testable question. Identify the independent, dependent and control variables and choose apparatus with suitable resolution.
3. **Risk assessment:** list the hazards of each chemical and procedure and how you will reduce the risk.
4. **Trial and refine:** carry out a preliminary run to choose sensible quantities and sampling times.
5. **Carry out:** e.g. drop a tablet into the acid at 37 °C, withdraw samples at timed intervals, and titrate each with $\text{KMnO}_4(\text{aq})$ to find how much Fe^{2+} has dissolved (for a KMnO_4 titration, model the stomach acid with dilute H_2SO_4 rather than HCl, because MnO_4^- also oxidises Cl^- ions; or use colorimetry instead). Repeat to check reproducibility.
6. **Evaluate and reference:** compare your findings with published information and cite every source you used in a consistent referencing style.

Safety

- Your own written risk assessment must be checked by your teacher before you start.
- Dilute HCl and KMnO_4 are irritants — goggles; KMnO_4 is an oxidiser and stains.
- Iron tablets must not be taken internally or removed from the lab.

Analysis

Process data: plot the amount of Fe^{2+} released against time and find when release is complete (or 50% complete). Comment on the uncertainty in each measurement.

Referencing: for each source give the author(s), title, publisher or website, year and (for web sources) the URL and date accessed. Judge reliability: is it peer-reviewed, recent, and from a reputable organisation?

Error Analysis

Source of Error	Type	Improvement
Model does not match real conditions (no stirring/enzymes)	Systematic	Stir at a constant rate; state the limitations of the model
Fe^{2+} oxidised to Fe^{3+} by air before titration	Systematic	Titrate samples immediately; keep them acidified
Tablets vary in mass/coating	Random	Use several tablets from the same batch and average
Unreliable or uncited sources	—	Use reputable sources and reference all of them

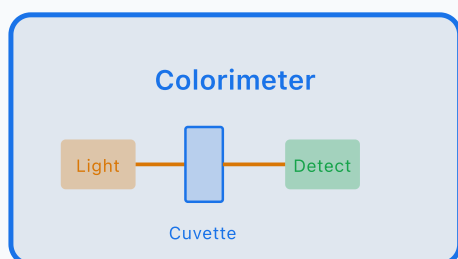
Exam Tip: Written papers can test PAG12 skills through planning questions: expect to write a method with named apparatus and quantities, identify variables to control, suggest a risk and its control measure, and evaluate whether a secondary source is reliable. Always justify your choices rather than just listing them.

Supporting Practicals (not PAGs)

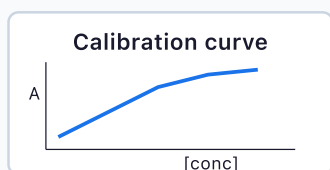
The three practicals below are **not** Practical Activity Groups in the OCR A specification. They are kept here because they practise examinable techniques: colorimetry can be used as a continuous-monitoring

method (PAG9), thin-layer chromatography is one of the PAG6 techniques, and the esterification equilibrium uses PAG2 titration skills.

Supporting: Colorimetry and Transition Metal Complexes



Absorbance: 0.45



Transition Metal Complexes

$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ — pale blue

$[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ — deep blue

$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ — violet (often looks green)

$[\text{Cr}(\text{NH}_3)_6]^{3+}$ — purple

$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ — pink

$$A = \epsilon cl$$

A = absorbance

ϵ = molar absorptivity

c = concentration, l = path length

Apparatus

- Colorimeter with appropriate colour filter, cuvettes
- Standard solutions of transition metal complex (various known concentrations)
- Test tubes for ligand substitution reactions
- Solutions: CuSO_4 , $\text{NH}_3(\text{aq})$, $\text{NaOH}(\text{aq})$, HCl , EDTA

Method

1. **Colorimetry:** Prepare a series of standard solutions of known concentration.
2. Select the complementary colour filter on the colorimeter (e.g., red/orange filter for blue Cu^{2+} solutions).
3. Zero the colorimeter using distilled water as the blank.

4. Measure the absorbance of each standard. Plot a calibration curve: absorbance vs. concentration.
5. Measure the absorbance of the unknown solution. Read the concentration from the calibration graph.
6. **Ligand substitution:** Add NH_3 dropwise to $\text{CuSO}_4(\text{aq})$. Observe colour change from pale blue to deep blue as NH_3 replaces H_2O ligands. Record observations for each drop added.

Safety

- NH_3 has irritant fumes — work in well-ventilated area.
- Transition metal salts can be harmful — wear gloves.
- Handle cuvettes by the top (fingerprints affect readings).

Analysis

Plot: Calibration curve of absorbance vs. $[\text{Cu}^{2+}]$. Should be linear (Beer-Lambert Law: $A = \epsilon cl$).

Calculate: Unknown concentration by interpolation from the calibration graph.

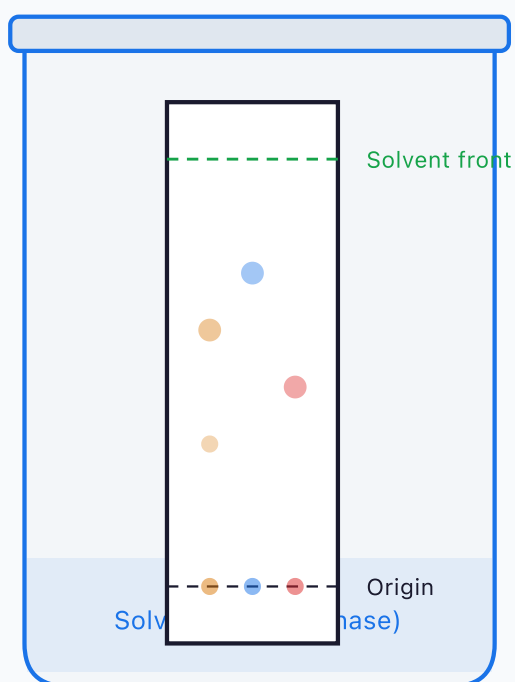
Error Analysis

Source of Error	Type	Improvement
Fingerprints on cuvette	Systematic	Handle by top; wipe with lens tissue
Wrong filter selected	Systematic	Use complementary colour filter; determine λ_{max}
Air bubbles in cuvette	Random	Tap gently to dislodge; fill carefully
Dilution errors in standard solutions	Random	Use volumetric flasks and pipettes

Exam Tip: The filter used in the colorimeter must be the complementary colour to the solution being measured. A blue solution absorbs orange/red light, so use an orange filter. The relationship $A = \epsilon cl$ (Beer-Lambert Law) is only valid at low concentrations — at high

concentrations, the calibration curve deviates from linearity. Always include a blank (distilled water) to zero the instrument.

Supporting: Thin-Layer Chromatography



R_f Value

$$R_f = \frac{\text{distance by spot}}{\text{distance by solvent}}$$

R_f is constant for given
solvent + substance

Key points:

- Pencil line (not ink)
- Solvent below origin
- Lid prevents solvent evaporation

Apparatus

- TLC plates (silica or alumina on plastic/glass backing)
- Developing chamber (beaker with lid/cling film)
- Capillary tubes (for spotting), pencil, ruler
- Suitable solvent or solvent mixture (mobile phase)
- UV lamp or iodine vapour (for visualising colourless spots)

Method

1. Draw a faint pencil line (origin) about 1 cm from the bottom of the TLC plate.
2. Spot samples onto the origin using capillary tubes. Keep spots small and evenly spaced. Include known reference compounds.
3. Pour solvent into the developing chamber to a depth of about 0.5 cm (below the origin line). Allow atmosphere to saturate.
4. Place the plate in the chamber. Ensure the origin line is above the solvent level.
5. Cover and allow the solvent to rise up the plate by capillary action.
6. When the solvent front nears the top (~1 cm from edge), remove the plate and immediately mark the solvent front with pencil.
7. If spots are colourless, visualise under UV lamp or in iodine vapour.
8. Measure distances from origin to centre of each spot and to the solvent front.
9. Calculate $R_f = \text{distance by spot} / \text{distance by solvent front}$.

Safety

- Many solvents are flammable and volatile — no flames; fume cupboard.
- UV lamp — do not look directly at the UV light.
- Iodine vapour is irritant — use in fume cupboard.

Analysis

Calculate: R_f values for each spot. Compare with reference values or known standards run alongside.

Identify: Components with the same R_f as known compounds (in the same solvent system) are likely the same substance.

Error Analysis

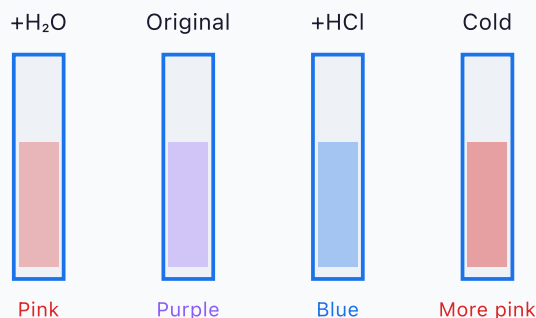
Source of Error	Type	Improvement
Spots too large (poor resolution)	Systematic	Apply small, concentrated spots; dry between applications

Solvent level above origin line	Systematic	Check depth before inserting plate
Solvent front not marked immediately	Random	Mark immediately on removal — solvent evaporates quickly
Chamber not saturated with vapour	Systematic	Line chamber with filter paper; allow time to saturate
Using ink pen instead of pencil	Systematic	Always use pencil — ink will dissolve in the solvent

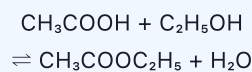
Exam Tip: R_f values are only valid if compared under identical conditions (same solvent, temperature, plate type). A substance is identified if its R_f matches a known reference run alongside it in the same experiment. If an unknown sample separates into two spots, it contains at least two components. TLC is a form of adsorption chromatography — polar substances adsorb more strongly to the polar silica stationary phase and have lower R_f values. Non-polar substances travel further (higher R_f).

Supporting: Investigating Equilibria

Le Chatelier's Principle



K_c Determination



Titrate with NaOH to find [acid] at equilibrium

$$K_c = \frac{[\text{ester}][\text{water}]}{[\text{acid}][\text{alcohol}]}$$



Only temperature changes K_c. Concentration shifts position but not K_c.

Apparatus

- Test tubes, water baths (ice and hot)
- CoCl₂ solution, conc. HCl, distilled water
- **For K_c**: Sealed tubes with ethanoic acid + ethanol + catalyst (H₂SO₄), NaOH for titration, burette, pipette, indicator

Method

1. **Le Chatelier's (qualitative)**: Prepare cobalt equilibrium mixture (purple). Add HCl → blue. Add water → pink. Heat → blue. Cool → pink.
2. **K_c determination (quantitative)**: Seal mixtures of ethanoic acid and ethanol with catalyst in tubes. Leave at constant temperature until equilibrium (several days or heat to speed up).
3. Titrate the equilibrium mixture with standardised NaOH to find moles of acid remaining.
4. Use ICE table: calculate moles of each species at equilibrium.
5. Calculate $K_c = \frac{[\text{products}]}{[\text{reactants}]}$.

Safety

- CoCl₂ is toxic — gloves.
- Conc. HCl — fume cupboard.

- Conc. H_2SO_4 catalyst — very corrosive.

Analysis

Qualitative: Record colour changes and link to Le Chatelier's principle.

Quantitative: ICE table to calculate equilibrium concentrations. $K_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}]}$.

Error Analysis

Source of Error	Type	Improvement
Equilibrium not reached	Systematic	Allow sufficient time; use catalyst; check with repeat
Temperature not constant	Systematic	Use thermostatted water bath
Titration disturbs equilibrium	Systematic	Quick titration; acknowledged limitation
Catalyst (H_2SO_4) also titrated by NaOH	Systematic	Run a blank with just H_2SO_4 to subtract; account for in calculation

Exam Tip: For the esterification K_c : remember to subtract the moles of H_2SO_4 catalyst from the titre (it also reacts with NaOH). The value of K_c for this reaction is approximately 4 at room temperature. A catalyst does not change K_c or the equilibrium position — it only speeds up attainment of equilibrium. When describing Le Chatelier's shifts, always state: (1) what change was made, (2) which direction the equilibrium shifts, (3) why (to counteract the change), and (4) what observable effect this has (colour change).

Worked Example — K_c from Esterification:

1.00 mol ethanoic acid and 1.00 mol ethanol were sealed with H_2SO_4 catalyst. At equilibrium, titration with NaOH showed 0.333 mol acid remaining (after subtracting the catalyst contribution).

Moles reacted = $1.00 - 0.333 = 0.667$ mol. From stoichiometry, moles of ester = 0.667, moles of water = 0.667, moles of alcohol = $1.00 - 0.667 = 0.333$.

$$K_c = (0.667 \times 0.667) / (0.333 \times 0.333) = 0.445 / 0.111 = \mathbf{4.01} \text{ (no units — moles cancel)}$$